

Avishkar Pawar

AI/ML Engineer

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SUMMARY

Computer Engineering student focused on AI/ML infrastructure, vector search, and applied machine learning. Built VectorFlow, an AI vector database with HNSW approximate nearest neighbor search, optimized distance computation, WAL persistence, and RAG pipeline integration. Also developed deep learning and computer vision projects, including retinal image disease detection and player re-identification. Completed Software Developer internships at OSMOS and Core-Decimal Solutions.

WORK EXPERIENCE

Software Developer Intern, OSMOS

Jul 2025 - May 2026

- Developed backend services using Node.js and Python for application and product development.
- Contributed to product development from scratch by designing, implementing, testing, and improving core features.

Software Developer Intern, Core-Decimal Solutions

Feb 2025 - Aug 2025

- Built full-stack features using Vue.js, Express.js, Sequelize, and MySQL in an MVC architecture.
- Developed reusable Vue.js components, integrated RESTful APIs with Axios, and designed Sequelize data mappings.

Software Development Intern, Campus Credential

2024

- Worked on web application features, database-backed workflows, and production-oriented bug fixes.

PROJECTS

VectorFlow - AI Vector Database | *Python, FastAPI, NumPy, Numba, LangChain*

[GitHub](#)

- Built HNSW nearest-neighbor search, WAL persistence, distributed sharding, REST APIs, optimized distance computation, and RAG pipeline integration for semantic search.

Job Search Copilot | *Python, Groq AI, automation*

[GitHub](#)

- Built a read-only assistant that searches job platforms, scores role fit with Groq AI, and writes ranked daily reports without auto-applying.

Player Re-Identification in Sports Footage | *Python, computer vision, deep learning*

[GitHub](#)

- Built a system for re-identifying players across camera angles in sports footage using visual features and tracking-oriented matching.

Eye Disease Detection Using Deep Learning | *Python, TensorFlow, Flask, OpenCV*

[GitHub](#)

- Developed a CNN-based retinal image classification workflow for detecting eye disease categories and serving predictions through a web interface.

EDUCATION

B.Tech. in Computer Engineering, MIT Academy of Engineering, Pune

2022 - 2026

CGPA: 9.24/10

ACHIEVEMENTS

- GATE 2026 Qualified: AIR 3124, GATE Score 601, 51.45/100 marks in Computer Science.
- GATE 2025 Qualified: AIR 9860, GATE Score 453, 39.32/100 marks.
- First Place, Technodium 25 technical competition.
- Certifications: AWS Cloud Practitioner, CCNA, Python Essentials, PCAP, NPTEL Modern C++, NPTEL Machine Learning, NPTEL DSA, NPTEL Cloud Computing, NPTEL Cryptography, Saylor Cryptography, Red Hat Academy.

SKILLS

Focus Areas AI/ML Infrastructure, Vector Search, RAG, Computer Vision, Python, FastAPI

Languages C++, Java, Python, JavaScript, TypeScript, SQL, PHP

Frameworks Spring Boot, FastAPI, React, Node.js, Express.js, Vue.js, Flask

Databases PostgreSQL, MySQL, MongoDB, Redis, custom database implementations

AI/ML TensorFlow, PyTorch, OpenAI API, LangChain, NumPy, Numba, computer vision

Tools Git, Docker, Linux, AWS, CMake, Maven, Prometheus, REST APIs

Core CS Data Structures and Algorithms, Operating Systems, Database Systems, Computer Networks, Distributed Systems, System Design